lab 3: familarization with NAND, NOR, XNOR and XOR Gate

Objective:

- To understand the principle and operation of NAND, NOR gates used in digital system.
- To implement, test and verify NAND, NOR, OR and XNOR logic Gates.

Theory:

NAND Gate

Truth table for NAND Gate

Input		Output
A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

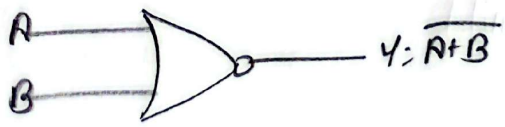
- Nand Gate is electronic circuit which Generate low output if both inputs are high otherwise high output.
- It is complement of NAND gate

NOR Gate

Truth table

Input		Output
A	B	$F = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

- NOR gate is electronic circuit which generate high output if both of inputs are low otherwise low output.
- It is complement of NOR Gate



OBSERVATIONS

XOR Gate

XOR Gate is an electronic circuit which produces low output when all inputs are same otherwise it produces high output. XOR gate is also known as exclusive -OR $F = A\bar{B} + \bar{A}B = A \oplus B$

Truth table

A	B	Expected Output	Observed Output
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

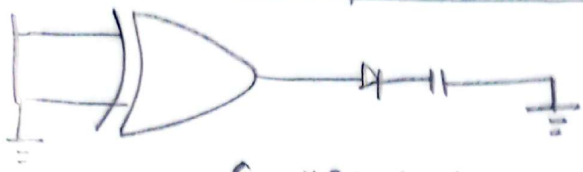


fig: XOR Gate

XNOR Gate

XNOR Gate is an electronic circuit which produces high output when all inputs are same otherwise it produces low output.

$F = AB + \bar{A}\bar{B} = A \odot B$

Truth table

A	B	Expected Output	observed output
0	0	1	1
0	1	0	0
1	0	0	0
1	1	1	1

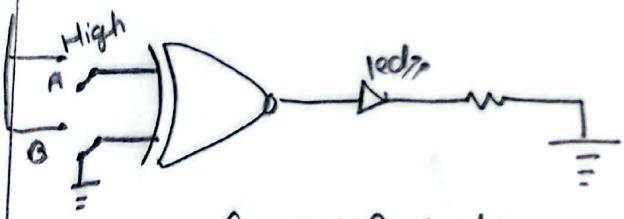
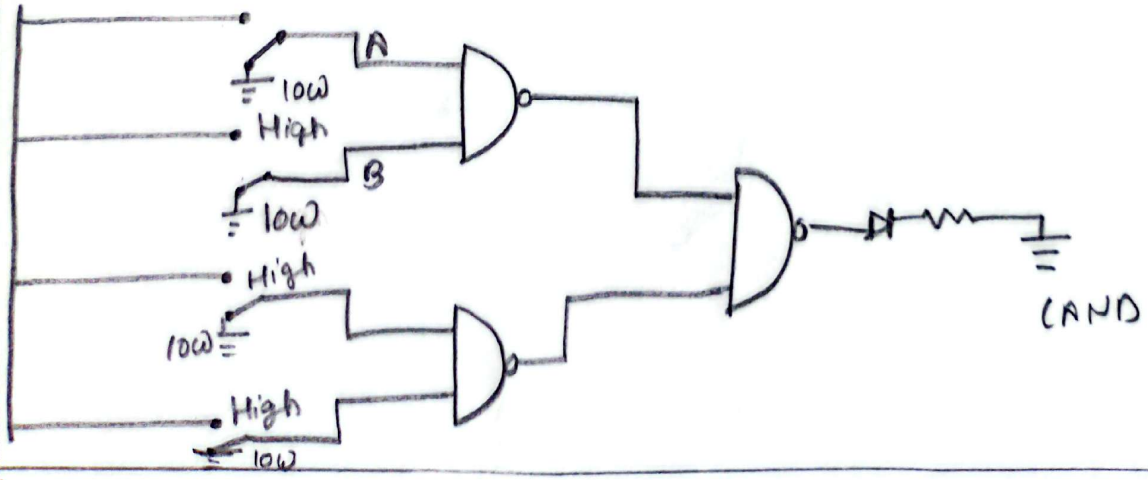


fig: XNOR Gate

Four Input NAND Gate

A four Input NAND gate is a digital logic gate that has four input terminals and one output terminal.



Truth table:

A	B	C	D	$\overline{AB}$	$\overline{CD}$	Expected Output	Observed
0	0	0	0	1	1	0	0
0	0	0	1	1	1	0	0
0	0	1	0	1	1	0	0
0	0	1	1	1	0	1	1
0	1	0	0	1	1	0	0
0	1	0	1	1	1	0	0
0	1	1	0	1	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	0
1	0	1	0	1	1	0	0
1	0	1	1	1	0	1	1
1	1	0	0	0	1	0	0
1	1	0	1	0	1	0	0
1	1	1	0	0	1	0	0
1	1	1	1	0	0	1	1

Conclusion

In this lab session we learned about various gates including NAND, NOR, XOR and XNOR gate. We are observed 3 level NAND or NOR circuit containing four inputs. In conclusion, we understood the principles and operation of NAND, NOR, XOR and XNOR gates.

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